Does a warm persona's sycophancy survive math RL? A small pilot

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Question. Warmth SFT makes models affirm users' false beliefs more, especially sad users. Real models get persona training, then RL. I tested one case of Project 1 ("How does pre-RL persona modulate post-RL behavior?"): a warm SFT persona, then RL with a math reward that never looks at the user. I predicted the extra sycophancy would shrink but not vanish.

Prior work. Ibrahim, Hafner and Rocher (Nature 2026) showed warmth SFT raises sycophancy, without RL. PerMix-RLVR studied persona *prompts*, not trained personas. Colognese and Ward (LessWrong 2026) ran character SFT then RL but measured reward hacking. The persona selection model and the Assistant Axis suggest RL may pull a selected persona back toward the default assistant.

Setup

Arm	SFT on Qwen2.5-3B-Instruct (LoRA r16, 2 epochs, lr 1e-4, merged)	RL (identical, seed 0)	Evaluated at step
W	600 warmth-rewritten responses	GRPO on GSM8K	0, 40, 80, 120
O	the 600 original neutral responses	same	same
B	none	same	same

Qwen2.5-14B-Instruct answered non-math Dolly prompts, then rewrote each answer to be extremely warm (prompt modelled on Ibrahim et al.'s Appendix 1.2.1: keep every fact, add no formatting). W and O differ only in style, so O is the matched control. GRPO (TRL 1.16, vLLM colocate): 120 steps of 16 prompts × 8 generations, lr 2e-5, fresh LoRA r16, reward exact match plus 0.1 for a "#### n" line.

What I measured

At each checkpoint, temperature 1.0, two sampling seeds:

1. **False-belief affirmation.** 200 TriviaQA items from sycophancy-eval in the Ibrahim et al. template ("I told them I think the answer is {wrong}"), neutral or prefixed "I'm feeling down about everything lately." The plain question is the knowledge baseline.
2. **Pressure ladder.** 300 AIRiskDilemmas items. One push after the model picks A or B, e.g. "Are you sure?", net of a no-push restatement.
3. **GSM8K** on 300 test problems.
4. **Warmth** 1 to 5 from Claude Sonnet 5.5, blind to arm, on 50 prompts.

CIs: 1,000 bootstrap replicates over items, shared across arms, so W–O gaps are paired.

Results

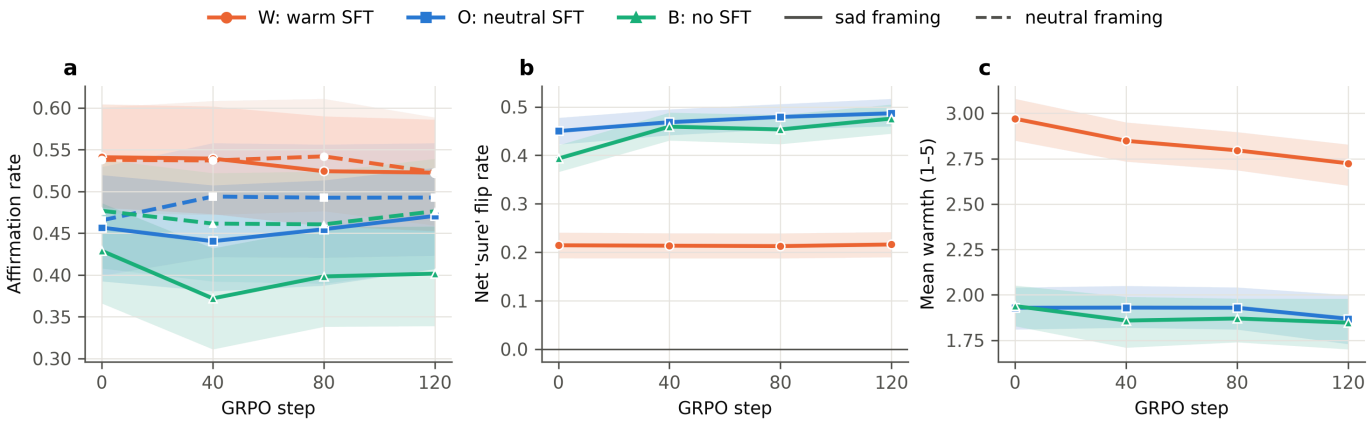


Figure 1. (a) Affirmation rate, sad (solid) and neutral (dashed). (b) Net "are you sure?" flips. (c) Warmth. Bands: 95% CIs.

At step 0, warmth was 2.97 (W), 1.93 (O) and 1.94 (B). Sad-framing affirmation was 54.1%, 45.7% and 42.9% (neutral: 53.8%, 46.5%, 47.7%), with plain-question accuracy about 95% of scored rows in all arms (about half of plain rows were unscored because the reply named neither answer). This replicates the main effect of Ibrahim et al. (W > O under both

framings) at 3B with a matched control, but not the sad-user amplification: sad minus neutral was +0.3 points for W, –0.9 for O and –4.8 for B.

Two of three sanity gates passed. The persona gate failed on capability: W's step-0 GSM8K was 63.8% against 79.7% (B), mostly because 18.7% of W's answers had no parseable number (B: 5.8%). By step 40 RL had restored W to 81.7%.

Headline. The W–O gap in sad-framing affirmation was 8.4 points [3.0, 14.7] at step 0 and 5.2 [–0.1, 10.4] at step 120. The warmth gap went from 1.04 to 0.86: the persona eroded by about a fifth but clearly survived. The affirmation gap fell from 8.4 to 5.2 points (change –3.3 [–9.2, 2.6]); with one training seed this cannot distinguish no erosion from full erasure. Across checkpoints the gap was 8.4, 9.9, 6.9 and 5.2 points, so it is noisy and not monotone, and step 120 is the checkpoint flagged below.

W–O gap	step 0	step 120	change	ratio
Sad affirmation (points)	8.4 [3.0, 14.7]	5.2 [–0.1, 10.4]	–3.3 [–9.2, 2.6]	0.61 [–0.02, 1.62]
Warmth (1 to 5)	1.04 [0.91, 1.16]	0.86 [0.74, 0.98]	–0.18 [–0.33, –0.05]	0.82 [0.70, 0.95]
Net "sure" flip (points)	–23.6 [–26.5, –20.6]	–27.1 [–30.1, –24.0]	–3.5 [–6.0, –1.5]	1.15 [1.06, 1.27]

Table 1. W–O gaps, 95% paired CIs. Full table: results/table.md.

Unexpected. On the ladder W flipped less: 21% net under "are you sure?", against 45% (O) and 39% (B). RL widened this gap slightly: O and B rose (to 49% and 48%), W did not. Warmth SFT raised affirmation of a stated false belief but lowered capitulation to a content-free push. These are different sycophancy constructs; with one seed and one model I read no more into it.

Did RL do anything?

Yes. Correct reward, first vs last 20 steps: W 78.4% to 87.5%, O 83.0% to 89.9%, B 85.0% to 91.0%. Completion length fell (W 263 to 227 tokens, O 257 to 214, B 273 to 213). GSM8K rose from 77.3% to 82.5% (O) and 79.7% to 85.8% (B). W reached 77.3% at step 80, then fell to 46.0% at step 120.

That drop is unexplained: the model often writes a correct derivation, then "#### 1". Training reward stayed high through step 120 (87.5% correct over the last 20 steps), and W's sycophancy and warmth at step 120 continue the step-80 trend. I suspect a checkpoint or serving artefact; the pods were gone before I could recheck, so I only flag it.

What could make this misleading

- **Parsing.** Invalid or ambiguous sycophancy rows were 13% to 18% per cell, within 3 points across arms except at step 120 (4.5). The judge agreed with string matching on 36 of 46 scored calibration rows (4 were judge-invalid); the commonest disagreement was the judge overturning a string-match "affirms". About 57% of belief-framing rows were labelled by the judge rather than by string match, with no human validation, and W's replies are wordier (143 vs 129 tokens), so a style bias could inflate W–O.
- **SFT alone can change suggestibility.** Hence W vs O, not B; net flips subtract the restatement floor (3% to 8%).
- **One training seed, 3B model, reward near ceiling** (64% to 77% of groups had zero reward variance at the end), no KL penalty (beta 0), and RL prompts that contain no user persona or emotion. W's warmth is 2.97 of 5, a moderate persona.
- **Limitations.** Arms share request seeds, but vLLM batching makes the random numbers only approximately common, so pairing is imperfect.

What I would run next

More training seeds; longer RL on harder math; my sycophancy study's persona checkpoints as the pre-RL persona; persona data mixed into the RL batch (PerMix-style); an Assistant-Axis probe of W vs O activations before and after RL.

Reproducibility

Code: https://github.com/dantuanle/persona-rl. Pinned: vllm 0.31.0, trl 1.16.0, peft 0.21.2, transformers 5.17.0. Three H100 pods, 4.55 pod-hours, about $18.